

Muhammad Karrar Ahmad

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PROFESSIONAL SUMMARY

Driven T-Level Digital Production student and Data Engineering Intern with hands-on experience designing automated data pipelines, replacing legacy Excel-based systems, and delivering scalable, data-driven solutions for real-world clients including Heathrow Airport. Proficient in Python, SQL, React.js, TypeScript, FastAPI, and MongoDB, with a strong ability to translate complex business requirements into efficient technical systems. Seeking a junior data or software engineering role where analytical thinking, technical skills, and initiative can deliver immediate impact.

KEY PROJECTS

Shift Management System - Ethos Farm | *React.js, JavaScript, SQL, HTML, CSS*

- Designed and built a full-stack shift management system replacing manual Excel-based scheduling for 400+ airport terminal staff across multiple locations
- Implemented a three-tier role architecture: Admin (system configuration, location and user management), Manager (absence monitoring, Return to Work notifications, data pipeline reporting), and Supervisor (7-day shift visibility, real-time staff assignment, sick leave marking)
- Supervisors can view upcoming staff rosters for the next 7 days, mark team members as absent or on sick leave, and track shift coverage in real time
- Managers receive live absence data, reasons for absence, and automated Return to Work (RTW) workflow notifications replacing a previously manual paper-based process
- Admins manage location configuration, user access controls, and system-wide settings through a dedicated admin panel
- Eliminated manual Excel roster uploads, significantly reducing administrative overhead and improving operational visibility across the organisation

Local Farmer E-Commerce Platform | *React.js, JavaScript, MongoDB, Python*

- Independently designing and building a full-stack e-commerce platform connecting local farmers directly with local buyers, eliminating intermediaries and supporting community agriculture
- Built as a T-Level Digital Production exam project, demonstrating end-to-end product development from requirements gathering through to implementation
- Features include product listings, buyer search and filter, order management, and farmer profile pages built with React.js frontend and Python/MongoDB backend
- Applying data-driven design principles to optimise the user experience and support inventory management for farmers with varying levels of technical literacy

Data Pipeline Automation | *Python, Pandas, SQL*

- Built automated ETL pipelines to collect, process and clean operational data from multiple sources, increasing accuracy and reducing reporting cycle time across 3 operational functions

PROFESSIONAL EXPERIENCE

Data Engineering Intern | *Ethos Farm*

Sep 2024 – Present

- Designed and delivered a digital data management system that eliminated legacy Excel-based workflows, reducing manual data entry time by approximately 60% across operational teams
- Engineered automated data pipelines using Python, Pandas and SQL, increasing data accuracy and reducing reporting cycle time across 3 operational functions
- Built a real-time shift management system supporting 400+ airport terminal staff with three-tier role architecture replacing manual roster uploads
- Led data collection, pipeline design and stakeholder reporting for a live feasibility study with Heathrow Airport, producing structured analysis used in infrastructure decision-making
- Collaborated with stakeholders to gather business requirements and translate them into scalable, production-ready technical solutions

- Project Lead** | *Fujitsu Sustainability Hackathon - Heathrow*

Jan 2025
- Led a cross-functional team of 5 to design a data-driven waste management solution for Heathrow Airport, securing 2nd place
 - Presented findings to industry judges using data visualisation and structured analysis
- Pathways Work Experience** | *Cisco - Remote*

Feb 2025
- Completed Cisco's Virtual Work Experience Programme, gaining practical skills in networking, cybersecurity fundamentals and collaborative problem-solving

PART-TIME EXPERIENCE

- Crew Member** | *McDonald's - Hayes*

Nov 2024 – Present
- Analysed shift-based operational data to identify service bottlenecks, proposing a staffing adjustment that improved peak-hour order speed by 15%
 - Recognised for consistent leadership and performance awarded Employee of the Month and Star of the Shift on 10 separate occasions
- Sous Chef & Hospitality Staff** | *Saffron D'Or / Manjaros*

Dec 2022 – Nov 2024
- Implemented a FIFO inventory control system with systematic data tracking, achieving a 15% reduction in material waste and measurable cost optimisation
 - Monitored competitor trends and industry data to inform seasonal menu decisions and event scheduling

ENTREPRENEURIAL EXPERIENCE

- Founder & Operator** | *E-Commerce Business — Pakistan*

2023 – Present
- Founded and operate an e-commerce business selling fashion, perfumes, jewellery and handbags, managing all end-to-end operations
 - Applied data-driven strategies to automate inventory reconciliation, digital marketing optimisation and online order management
 - Built a scalable business model with a clear roadmap for international expansion and worldwide shipping

EDUCATION

- T-Level — Digital Production, Design & Development** | *Uxbridge College, Hillingdon*

Sep 2024 – Present
- Equivalent to 168 UCAS points (3 A-Levels) | Achieved Grade B in First Year
 - Exam project: Local Farmer E-Commerce Platform — full-stack web application built independently using React.js, MongoDB and Python
- GCSE Maths (4) · English (5) · BTech Level 2 ICT (Double Merit)** | *West London College, Ealing*

Jun 2024
- Google Data Analytics Professional Certificate** | *Google / Coursera*

In Progress
- Foundations: Data, Data, Everywhere (Dec 2025) — [View Certificate](#)
 - Ask Questions to Make Data-Driven Decisions (Dec 2025) — [View Certificate](#)
- ESOL Level 1** | *Hammersmith College*

May 2023
- ICS - Information of Computer Science (Grade C)** | *Aspire College, Lahore*

May 2022
- Matriculation - Information Technology (A*)** | *Muslim Scholar Model High School, Lahore*

May 2021

SKILLS

Data Engineering: Python, Pandas, SQL, ETL Pipelines, Data Cleaning, Data Modelling, Excel Migration
Software Development: React.js, JavaScript, TypeScript, FastAPI, MongoDB, HTML, CSS, REST APIs, Git
Analytical: Exploratory Data Analysis, Process Optimisation, Data Quality, Stakeholder Reporting

Visualisation: Matplotlib, Seaborn, Dashboard Design

Leadership & Soft: Project Management, Stakeholder Communication, Team Leadership, Pitching & Presentation

Other: Cybersecurity Fundamentals, Networking Basics, System Architecture

Languages: English (Professional Proficiency) | Urdu (Native) | Punjabi (Native)

CERTIFICATIONS

- IT Work Experience Programme | Heathrow Airport | Apr 2024 — [View Certificate](#)
- Pathways Virtual Work Experience | Cisco | Feb 2025
- Professional Certificate in Data Analysis | Udemy | Jun 2025 — [View Certificate](#)
- Foundations: Data, Data, Everywhere | Google / Coursera | Dec 2025 — [View Certificate](#)
- Ask Questions to Make Data-Driven Decisions | Google / Coursera | Dec 2025 — [View Certificate](#)

INTERESTS

Badminton | Snooker | Gym | Photography | E-Commerce